

Note that in the above, we have used both

$$\frac{1}{2}\Sigma^3 u_+^k(0) = \pm \frac{1}{2}u_+^k(0) \quad \& \quad \frac{1}{2}\Sigma^3 u_-^j(0) = \pm \frac{1}{2}u_-^j(0)$$

which comes from the following list of facts:

$$\begin{aligned} \text{(i)} \quad & \frac{1}{2} \begin{pmatrix} \sigma^3 & 0 \\ 0 & \sigma^3 \end{pmatrix} u_+^1(0) = \frac{1}{2} \begin{pmatrix} \sigma^3 & 0 \\ 0 & \sigma^3 \end{pmatrix} \begin{pmatrix} \varphi_+^1 \\ \varphi_+^1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \varphi_+^1 \\ \varphi_+^1 \end{pmatrix} \\ \text{(ii)} \quad & \frac{1}{2} \begin{pmatrix} \sigma^3 & 0 \\ 0 & \sigma^3 \end{pmatrix} u_+^2(0) = \frac{1}{2} \begin{pmatrix} \sigma^3 & 0 \\ 0 & \sigma^3 \end{pmatrix} \begin{pmatrix} \varphi_+^2 \\ \varphi_+^2 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} \varphi_+^2 \\ \varphi_+^2 \end{pmatrix} \\ \text{(iii)} \quad & \frac{1}{2} \begin{pmatrix} \sigma^3 & 0 \\ 0 & \sigma^3 \end{pmatrix} u_-^1(0) = \frac{1}{2} \begin{pmatrix} \sigma^3 & 0 \\ 0 & \sigma^3 \end{pmatrix} \begin{pmatrix} \varphi_-^1 \\ \varphi_-^1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} \varphi_-^1 \\ \varphi_-^1 \end{pmatrix} \\ \text{(iv)} \quad & \frac{1}{2} \begin{pmatrix} \sigma^3 & 0 \\ 0 & \sigma^3 \end{pmatrix} u_-^2(0) = \frac{1}{2} \begin{pmatrix} \sigma^3 & 0 \\ 0 & \sigma^3 \end{pmatrix} \begin{pmatrix} \varphi_-^2 \\ \varphi_-^2 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} \varphi_-^2 \\ \varphi_-^2 \end{pmatrix} \end{aligned}$$